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Framework for Voice-controlled AI Teaching Assistant for Specially-abled

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***Abstract*—Recent advancements in technology have triggered a revolution in the education sector. One of the significant developments of this change is being granted access to vast online resources from various world class institutes. The motivation behind this paper is to be able to create a technological solution for providing equal opportunities to specially-abled online learners who may have physical disabilities such as visual impairment, loss of hands/weak hand motor skills, etc., and enable their participation for a more inclusive society. The paper uses a two-fold approach. Firstly, it provides literature on various state-of-art assistive technologies for e-learning portals. Secondly, the paper proposes a learning framework as teaching assistant for specially-abled, which is artificial intelligence based and interactive voice enabled. The proposed framework has three modules completely controlled by voice commands for self-paced - reading, learning and auto-marking assessment. The assessment module generates fill-in-the-blanks and question-answers, which is a part of continuous comprehensive tests. As a case study, the proposed framework is prototyped on an e-learning platform DIKSHA, which hosts digital books from National Council of Educational Research and Training. Natural language processing toolkit based on Python libraries has been used. The results have been found encouraging. Extensions for future work are also discussed.**

***Index Terms*—Natural Language Processing, E-Learning Portal, Voice-Based, Specially-abled, Inclusive Society**

I. Introduction

WE live in a digital era. Advancements in technology have seeped into various sectors of our life, and education has not been left behind. This revolution has eliminated the need to carry stacks of books to school, decreased the teacher's workload and has made learning more efficient. Massive Open Online Courses (MOOC) have transformed the education system, as its significant advantage includes both flexibility and affordability with unlimited participation - thus paving a path to career advancement. MOOC providers bring the best courses from the best schools (that includes reputed tech giants and world class academic institutes) at the learner's doorstep. Thus the learner's focus shift towards flexible schedule and self-learning at higher secondary and university levels. But at the same time, the specially-abled learners are not able to cope up with the same pace due to their physical limitations. The wide spectrum of disabilities include visually challenged, hand amputees, wheelchair users, weak hand motor skills, which hinder them from accessing the online resources. Research is going on to create technological solution for providing equal opportunities to the specially-abled, enhancing their experience using online e-learning platform and enable their participation in an inclusive society. A large percentage of such learners lack employment as the resources are not easily available, the number of special institutes available are not enough to educate such a large population. Lots of research are conducted under the umbrella of assistive technology over the decades, which include wearable devices and mobile applications [1], [2].

The evolution of Natural Language Processing (NLP) is seen as an emerging field to interact between computer and human language, combining the concepts from subjects such as linguistics, artificial intelligence and computer science. The packages and libraries related to NLP helps to create models that can communicate and respond like humans. There is a great demand to serve specially-abled in the education sector [3]–[5].

The paper focuses on providing a voice enabled interactive learning framework creating an enabling environment for specially abled learners for their empowerment based on natural language processing , a branch of artificial intelligence. The paper is divided into following sections. Section 2 provides literature of state-of-art assistive technologies for e-learning portals. Section 3 puts forth the proposed methodology of Voice-controlled AI Teaching Assistant Framework. Section 4 details out the prototyping experimentation and results. The paper concludes with future work in Section 5.

II. Related Work

The following sub-sections showcase the literature survey of the assistive technologies used for specially-abled learners, especially the visually impaired as end users.

A. Computer Vision for Assistive Technologies

The authors in [6], created a learning platform by combining deep convolutional networks (CNN), recurrent neural networks(R-CNN), and speech corpus models to provide an interactive learning experience for particular subjects. The authors in [7] designed an inexpensive wearable device that

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uses computer vision (CV) to read documents placed near to the user in various alignments and lighting conditions. The systems proposed by researchers in [8]–[10] use a combination of techniques such as machine learning, image processing and speech synthesis techniques.

B. Text-to-Speech and Speech-to-Text

In the paper [11] the author explains different approaches for text to speech synthesis, articular synthesis and formant synthesis.The author in [12] explains the working of speech to text conversion using Mel Frequency Cepstral Coefficient (MFCC) and support vector machine. The proposed model experimented on mix of couple of languages which includes English and Marathi, an Indian language of state of Maharashtra. The works proposed by authors in [13], [14], [15], [16] detail about using Digital Assistance for the Blind using the Text to Speech and Speech to Text libraries for tutoring.

C. Mobile Based AI Applications

As technology advanced and the use of mobile phones became very popular, the presence of innumerable applications surfaced up, serving various purposes for the visually impaired such as color detection [17], motion marking menus [18], image recognition [19], [20], Google Vision APIs, Voice Dream Reader.

D. Continuous Comprehensive Assessment (CCA)

Continuous assessment has become an integral part of any education system, which are usually short and concise in nature such as quiz, question and answers, fill in the blanks. Question and answers are mostly generated by the human experts in the respective subjects. There is a limitation in the number of human generated questions.

The author in [21] has built the model which is based on sentence selection, gap selection, distractor selection which includes semantic similarity, syntactic similarity, and contextual fit. Word2Vec library along with probabilistic functions is used to choose the blank generators in [22]. The model gave satisfactory results and the quality of the questions generated was evaluated. In paper [23], the author has used the word-embedding concept, sequence classification and sequence labelling. CoreNLP was used first to bring the sentences to word-level and thereafter for the sequence labelling and sequence classification. 2-layer bidirectional LSTM was applied, for training and evaluating the model using the data collected from user-interface. The authors in the paper [24], [25], [26] explain the concept of filling in the blanks and missing words in articles. Comparative study is done among the Bidirectional Encoder Representations from Transformers (BERT), ELMo (context setting word embedding method) and Temporal Convolutional Networks (TCN) model. In the paper [27], the author uses WordNet library. The paper [28] proposes the state-of-art model for the question and answers generation, using the multi-flow attention model, transformers, word-by-word generation. The author [29] has worked on a model for question generator. The paper [30] proposes a novel pre-training and tuning framework to generate and predict words closer to human writing patterns, which helps in achieving good state of art results in summarization, question answering, question generation.

To the authors best of knowledge, all these related works are not end-to-end voice assisted process for reading, learning and assessment taken together. The motivation of the paper is to develop a framework for reading, learning and assessment using voice assisted teaching assistant. Prototyping has been done for the school learners which features voice based easy selection of class, chapter and particular topic for reading and learning. The proposed methodology is detailed out in subsequent section.

III. Proposed Methodology

Most of the related work mentioned are proposed keeping the visually impaired. The authors want to address wide spectrum of specially abled which includes visually impaired and other disabilities which restrict the usage of using the e-learning platform due to poor hand motor skills as well. In National Education Policy NEP 2020 (https://www.education.gov.in//en//national-education-policy-2020-0), various new provisions for children with disabilities are encouraged to enable them to fully participate with regular schooling process.

The paper proposes a framework for Voice controlled AI Teaching Assistant built to mainly focus on helping specially-abled learners using online platform based on natural language processing. The authors have considered an existing online portal "DIKSHA" contents (easily accessible and freely downloadable) for prototyping the proposed methodology. DIKSHA is one of the online learning platform by India government for school learners (https://diksha.gov.in/ncert/). DIKSHA is developed on open source technology, made in India, which incorporates internet-scale technologies and enables various use-cases and solutions for teaching and learning. The platform is being leveraged and developed for foundational learning programs, school education and to support inclusive learning for differently-abled communities of learners and teachers [31]. In the context of COVID-19 related disruption of schooling, DIKSHA made it feasible for all states and union territories to enable education at home through innovative programs. The platform works very well for normal learners (without disability), but issues are being faced by the specially-abled students as per their parents reports. The article [32] "Visually impaired struggle with NCERT e-textbooks" published in a national newspaper in India, mentions the details of the issues faced by the specially-abled learners [33]–[35].

The strength of the proposed framework prototyping is that it does not need any supervision, specific dataset and covers holistically reading, learning, and assessing of any digital text content of the user's interest. In future the framework will be extended as a service plugged into any such e-learning portal. The framework has three modules : 1) Reading, 2) Learning and 3) Assessment as shown in Fig. 1. Firstly, the user needs

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to login to the DIKSHA portal and he/she selects the class, subject, chapter and after doing so, the user can choose any of the three modules from the framework. The user also has the option of choosing a language for any of these modules (by default, it is English). In future, the prototype can be extended to any language.

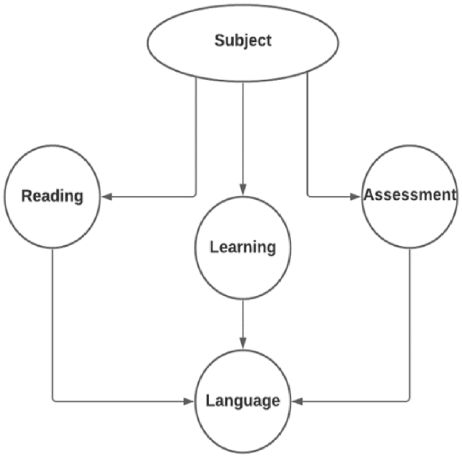


Fig. 1. Schematic Diagram of the Voice controlled AI Teaching Assistant

Detailed working explanation of the modules is discussed in the following subsections.

A. Voice Controlled AI Reading Module

The following pseudocode depicts step by step process of the reading module, selects learner's choice of subject and lesson in the preferred language.

```
1.START
2.IF Old user
3.   OUTPUT AND READ OUT \Do you want to
      continue from the last session?"
4.   IF(voice input=="yes")
5.      RETRIEVE the index of sentence list and
         paragraph list from last session
6.      REPEAT
7.      OUTPUT AND READ OUT  lesson
8.      IF the character is equal to \."
9.         ADD the sentence to sentence list
10.       Sentence list index++
11.    ELSE IF the character is equal to \  "
12. ADD the word to word list
13. Word list index++
14.    ELSE IF the character is equal to \\n"
15. ADD the paragraph to the para list
16.        Para list index++
17.  UNTIL voice input not equal to \stop"
18. ELSE
19 SET sentence list index=0, para list index=0 and
       word list index=0
20. OUTPUT AND READ OUT \select the subject"
21. VOICE INPUT subject
22. OUTPUT AND READ OUT \select the lesson"
23. VOICE INPUT lesson
24. GOTO 7
25.Else  GOTO 18
26.  IF (voice input == \repeat the last sentence")
27. Sentence list index--
28. GOTO 7
29.  ELSE IF(voice input == \repeat the paragraph")
30.      Para list index--
31.      GOTO 7
32.  ELSE IF(voice input == \Take test")
33.      CALL Take_test()
34.  ELSE IF(voice input == \Learning")
35.      CALL Learning()
36.  ELSE IF(voice input == \exit")
37.      GOTO END
38.END
```

The Table I provides information about the Python packages and functions used for the reading module.

The natural language processing libraries are extensively used for these modules. The model takes a word as input from the user and then the model gives some of the basic information about the word, such as synonyms, antonyms, word definition, an example sentence, parts of speech. Finally the user also has the option of getting the translation of that particular word in any preferred language. WordNet, a lexical database is used to get all the information of the words. It links words into semantic relations including synonyms, meronyms, and hyponyms. The synonyms are grouped into synsets with quick definitions and usage examples. Nouns, adjectives, verbs, and adverbs are assorted into sets of synonyms called synsets, each expressing a different concept.

The Table 2 provides information about the Python packages and functions used in the learning module of the AI based voice assisted teaching assistant. Next section describes the assessment module.

B. Voice controlled AI Assessment Module

The continuous comprehensive tests can be automated by creating quizzes including fill-in-the blanks, true or false, etc. The framework assessment module gives the learner the options of a) question and answer assessment b) fill in the blanks assessment, based on the choice of subject and the chapter. The methodology is totally unsupervised and is independent of any particular dataset.

Questions and Answers Assessment: The model prepares the questions from the given lesson. First, the sentences from the lesson are taken and then the sentences are further divided into words, then the parts of speech tagging is done onto those words. Constituency tree and the dependency graph of the sentences are drawn by the model, based on the main verb and main clause found in the sentence. Using the information of parts of speech and the information gathered from the trees, the 'Wh'word is applied in the sentence to generate the questions, such as 'What', 'Where', 'When', 'Who', etc. The questions are read out, and the user has to answer the questions, which will be recorded and then converted to text for further analysis and assessment.

Fill in the Blanks Assessment: In the present scope, the blanks are filled with nouns, can be either singular(NN), plural(NNS) or a proper noun(NNP). The appropriate noun is replaced with a blank. The sentences are spelt out for the user and the model waits for around 10 seconds to get the answer from the user. The answer is fed into the system in the form of voice and converted to text using the speech to text. The text is matched with the answer and upon matching the model increases the score count by appropriate value. The model spells out the final score without any human assistance.

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TABLE I
PYTHON FUNCTIONS USED FOR READING MODULE

SN	Package	Function	Description
1	nltk.tokenize	PunktSentenceTokenize()	PunktSentenceTokenizer is the abstract class for the default sentence tokenizer.
2	nltk	sent_tokenize()	Uses an instance of PunktSentenceTokenizer from the nltk.tokenize.punkt module, which has already been trained to mark the end and beginning of sentence
3	nltk.tokenize	tokenize ()	Provides a lexical scanner
4	Gtts	gTTs()	gTTS , a Python library and CLI tool to interface with Google Translate's text-to-speech API. Supports several languages including English, Hindi, Tamil, French, German, and many more.
5	playsound	playsound()	Cross-platform module that can play audio files
6	speech_recognition	Recognize(), listen()	Recognize and listen are used to get the voice input from the user, listen is used to set the time to record the input and also remove the background noises if any, while recognize is used to use the microphone of the system.
7	speech_recognition	recognize_google()	Recognize_google takes recorded speech as input and then gives the transcript of it.
8	nltk	pos_tag()	To tag the parts of speech of a given word.
9	nltk	word_tokenize()	To token the text which are words.

TABLE II
PYTHON FUNCTIONS USED FOR LEARNING MODULE

SN	Package	Function	Description
1	nltk.corpus	synsets()	Provides look-up words in WordNet which are groupings of identical words
2	nltk.corpus	lemmas()	To extract synonyms as well as antonyms of a particular synset from the wordnet.
3	nltk.corpus	definition()	Provides the definition of a given word
4	nltk.corpus	lemma().name()	Provides the synonyms
5	nltk.corpus	lemmas().antonyms().name()	Provides the antonyms of the word.
6	nltk.corpus	examples()	Provides the example sentences of a given word
7	Translator	translator()	Provides the translation of a word from one language to another

For prototyping, content is taken from Class 6 History book taught from DIKSHA. The parser transforms a sentence internally into dependency tree and a directed graph as shown in Figure 2. It depicts the dependency parser for a sentence.

```
det(empire-2, The-1)
nsubj:pass(founded-7, empire-2)
mark(ruled-5, that-3)
nsubj(ruled-5, Ashoka-4)
dep(empire-2, ruled-5)
aux:pass(founded-7, was-6)
root(ROOT-0, founded-7)
case(grandfather-10, by-8)
nmod:poss(grandfather-10, his-9)
obl:by(founded-7, grandfather-10)
compound(Maurya-13, Chandragupta-12)
appos(grandfather-10, Maurya-13)
advmod(2300-17, more-15)
fixed(more-15, than-16)
nummod(years-18, 2300-17)
obl:npmod(ago-19, years-18)
advmod(grandfather-10, ago-19)
```

The above relations give insights about the dependency type such as subject, determinant, modifiers,etc. A small subset of dependency relations are shown in Table III. The website (https://universaldependencies.org/u/dep/index.html)refers the entire list. The assessment module flowchart is shown in Fig.3.

TABLE III
FEW DEPENDENCY RELATIONS

Nominal Modifier Relations	Description
nmod	Nominal Modifier
nummod	Numeric Modifier
det	Determiner
appos	Appositional Modifier
advmod	Adverbial Modifier

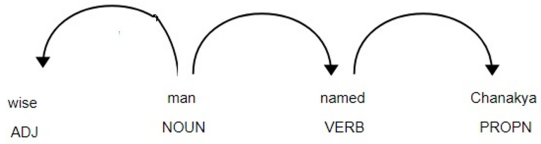


Fig. 2. Sentence Transformed to Dependence Tree

IV. EXPERIMENT AND RESULT ANALYSIS

The framework has been tested as a prototype case study on the DIKSHA portal. The experiments were conducted on intermediate physics, chemistry, biology and history subjects for school students of class 6,7,8 and 9.

A. Experimental Results for Voice Assisted Reading and Learning

The experiments are done for the reading module, where the model is given lessons from different subjects. The model

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read them well. The learner repeats the paragraphs as per requirement. The computer voice used as assistance was understood clearly by the learners (tested with 15 learners).To understand the vocabulary, the model produces synonyms, antonyms, definition, an example sentence, translated to a language of choice for better understanding. The NLTK corpus functions provided accurate results.

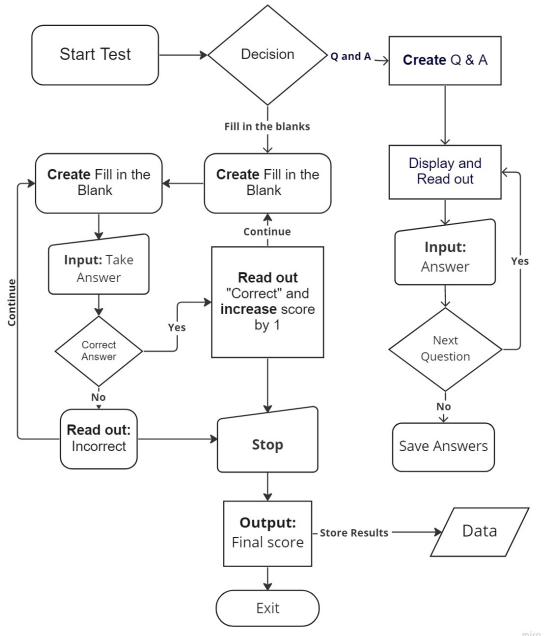


Fig. 3. Voice Controlled AI Assessment Module

B. Experimental Results of Voice Assisted Continuous Objective Assessment

The assessment module is bit complex and tricky. The semantic summarization of the chapter can be built in future work as an intermediary step for generating CCA questions and answers. The summarization in bulleted form can then be directly fed to this module to avoid all irrelevant questions generated due to meaningless formation of question making process. The model is given a paragraph from any subject to create fill in the blanks. At present the model considers the paragraphs directly. The user answers are received in voice mode and converted to text. The results for assessment criteria are displayed as shown in Table IV along with input given to the model. The human generated refers to teacher/human intervention.

TABLE IV
EXPERIMENTAL RESULTS OF FILL IN THE BLANKS ASSESSMENT

S.N	Subject	Human generated	Model Generated	Relevant/Total
1.	Physics	2	7	6/7
2.	Geography	4	15	14/15
3.	History	7	19	17/19

Model Limitation: Out of all the questions generated, some do not make sense. A semantic layer has to be incorporated in future for assessment and generation of image description. Training shall be considered as future work which will have subject and language dependency as well.

V. CONCLUSION AND FUTURE WORK

The paper proposes an artificial intelligence based voice enabled teaching assistant which can be integrated with the e-learning portals in future, to make the specially-abled learners and instructors fraternity to work independently. The paper provided a comprehensive literature on assistive technologies developed and researched on e-learning for specially abled, most of them focussed on visually impaired. To the authors best of knowledge, none of the related works are end-to-end voice assisted product which can be plugged to any of the MOOCs for reading, learning and assessment taken together for any learning context in general. The paper proposes a voice assisted framework to create a technological solution for providing equal opportunities to specially-abled online learners who may have physical disabilities such as visual impairment, loss of hands/weak hand motor skills, etc., and enable their participation for a more inclusive society. The three modules are based on natural language processing and speech processing. The reading module, learning module and assessment modules have been prototyped on one such publicly available online platform called DIKSHA, which hosts digital books from NCERT. The development is carried out using Python libraries for Natural Language Tool Kit (NLTK). The results are encouraging especially with no dependency on any special hardware, no dependency on screen readers, and without any supervised learning on any particular datasets. The proposed model is a simple to use and special-abled friendly. Finally, the model does almost all the work of human teaching right from reading to teaching, generating objective type assessment tests and finally automated assessment - totally voice assisted. The future work of this paper includes an improved version of assessment module with text summarization as the core engine. This will increase the efficiency of question generation. The reading of equations and images are still open areas of research. In the future, the improved version of the proposed framework will be extended as a service interface for any e-learning portal to provide an inclusive learning ecosystem.

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